



FACULTY OF COMPUTING
& INFORMATION TECHNOLOGY
KING ABDULAZIZ UNIVERSITY



كلية الحاسبات
وتقنية المعلومات
جامعة الملك عبدالعزيز

CPIT 251 - Project

(Medication Tracker)

Student Name	ID
1. Ali Salem Battiyah	2338097
2. Bandar al-Jedani	2237428
3. Ahmad Al-Ghamdi	2337548
4. Abdulaziz Al-Orabi	2339698
Section: IT2	

External Partner: Seniors, healthcare providers, caregivers.

Interview outline	
Interviewee: Older people, who are 50-60 years old and have medication Healthcare medication	Interviewer: Abdulaziz Alorabi Ali Battiyah Ahmad Alghamdi Bandar Al-Jedani
Location: Senior living faculty, at family member's home, on phone.	Appointment Date: Start time: 8:30 End time: 9:30
Objectives: The primary objective of the interview is to learn more about the difficulties older people, their caregivers, and healthcare professionals face when managing their medications. It also aims to assess their needs and collect input that will inform the Medication Tracker application's design and development.	Reminders: Ensure that the person you interview is comfortable. Don't ask irrelevant questions. Speak clearly and slowly. Record answers accurately, don't be shy to ask the same question again.
Question 1: How many medications do you usually take in a day, and at what times?	I take about 4 medications daily, one in the morning before I eat, one after lunch after I eat, one in the evening, and one at sleep.
Question 2: Do you sometimes forget to take your medication? If yes, how often does this happen?	Yes, I sometimes forget, especially in the afternoon when I wake up. It happens about 2 times a week. I think it's because I have some medications that I need to take before meals and others after meals, and I sometimes lose track of which ones I've already taken.
Question 3: What usually causes you to miss or delay a dose (e.g., being busy, forgetting, not hearing an alarm)?	I often forget because I'm busy with other things or miss hearing my alarm warning.
Question 4: How do you currently try to remember your medications (alarm clock, written notes, family reminders, etc.)?	I usually depend on my relatives reminding me when they come to visit and use a notebook with alarm.

Question 5: Would you like to receive reminders on your phone when it's time to take your medicine? Why or why not?	Yes, it could help with recollection, especially for the doses I sometimes miss.
Question 6: Would it be helpful if a family member or caregiver also got a notification when you miss a dose?	Yes, knowing that someone will be informed if I forget makes me feel secure.
Question 7: What would make a medication reminder app easier for you to use (large text, voice instructions, pictures of medication, etc.)?	It would be extremely helpful to have photos of the medication, big fonts, and instruction in a clear voice.
Question 8: Would you like an app to prepare a report on your medication history for your doctor?	Yes, since everything would be planned, going to the doctor would be simpler.
Question 9: For healthcare, what features would make such an app clinically reliable and easy for you to monitor patient adherence?	Real-time notifications of missed doses Ability to generate adherence reports Integration with electronic health records
Question 10: For healthcare, which types of medications are most often forgotten or mismanaged by seniors?	Daily pills like blood pressure and diabetes medication Medications taken multiple times per day Some medication taken weekly such as vitamin D

Functional Requirements

1. User Accounts

- The app should allow seniors and caregivers to register and log in.
- Caregivers should be able to connect to the senior's account.

2. Medication Input

- The user or caregiver should be able to add, edit, and delete medications.
- Each medication should include name, dosage, and schedule.
- The app should allow adding pill photos for easier identification.

3. Reminders and Notifications

- The app should send reminders (audio + visual) when it's time to take medicine.
- Voice guidance should be available to help seniors.
- The user should be able to confirm if the dose was taken or missed.

4. Tracking and Logging

- The system should keep a history of all taken and missed doses.
- It should generate reports that can be printed for doctors.

5. Caregiver Alerts

- If multiple doses are missed, caregivers should get an alert.
- Caregivers should be able to monitor the medication schedule.

6. Offline Mode

- The app should work even without internet (basic functions like reminders and logs).
- Data should sync once the internet is available again.

Non-Functional Requirements

1. Usability

- The app should have a simple and easy-to-use interface.
- Buttons should be large, and text should be high contrast for seniors.
- Voice guidance should help users with limited reading ability.

2. Accessibility

- The app should support elderly users by being senior-friendly.
- It should work for users with weak eyesight or hearing issues.

3. Reliability

- Reminders must always work at the correct time.
- The system should be available most of the time with minimal crashes.

4. Performance

- Notifications should not be delayed.
- The app should respond quickly (within a couple of seconds).

5. Security

- Users' medication and health data should be kept private.
- Log-in should be protected with a password.

6. Scalability

- The system should support multiple seniors and caregivers.
- It should allow future updates and extra features.

7. Maintainability

- The project code should be easy to understand and updated.
- Documentation should be provided for future improvements.